STACKS, DISCRIMINANTS: LEMMA-DUALIZING-TAU

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 997-1001 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1003-1104, the complete author proof (100 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_discriminant.json (pb=4592, stmt 997-1001, proof 1003-1104).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00EK: sources/stacks-discriminant/pinned-original-source.tex, source lines 1104–1104. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BTA: sources/stacks-discriminant/pinned-original-source.tex, source lines 955–961. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT9: sources/stacks-discriminant/pinned-original-source.tex, source lines 894–907. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT7: sources/stacks-discriminant/pinned-original-source.tex, source lines 843–846. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a flat quasi-finite map of Noetherian rings. There exists a trace element $\tau \in \omega_{B/A}$.*

Proof. Choose a factorization $A \to B' \to B$ with $A \to B'$ finite and $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ an open immersion. Let $g_1, \ldots, g_n \in B'$ be elements such that $\mathrm{Spec}(B) = \bigcup D(g_i)$ as opens of $\mathrm{Spec}(B')$. Suppose that we can prove the existence of trace elements τ_i for the quasi-finite flat ring maps $A \to B_{g_i}$. Then for all i, j the elements τ_i and τ_j map to trace elements of $\omega_{B_{g_i g_j}/A}$ by Lemma 0BT9. By uniqueness of

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trace elements (Lemma 0BT7) they map to the same element. Hence the sheaf condition for the quasi-coherent module associated to $\omega_{B/A}$ (see Algebra, Lemma 00EK) produces an element $\tau \in \omega_{B/A}$. Then τ is a trace element by Lemma 0BTA. In this way we reduce to the case treated in the next paragraph.

Assume we have $A \to B'$ finite and $g \in B'$ with $B = B'_g$ flat over A. It is our task to construct a trace element in $\omega_{B/A} = \mathrm{Hom}_A(B', A) \otimes_{B'} B$. Choose a resolution $F_1 \to F_0 \to B' \to 0$ of B' by finite free A-modules F_0 and F_1. Then we have an exact sequence

$$0 \to \mathrm{Hom}_A(B', A) \to F_0^\vee \to F_1^\vee$$

where $F_i^\vee = \mathrm{Hom}_A(F_i, A)$ is the dual finite free module. Similarly we have the exact sequence

$$0 \to \mathrm{Hom}_A(B', B') \to F_0^\vee \otimes_A B' \to F_1^\vee \otimes_A B'$$

The idea of the construction of τ is to use the diagram

$$B' \xrightarrow{\mu} \mathrm{Hom}_A(B', B') \leftarrow \mathrm{Hom}_A(B', A) \otimes_A B' \xrightarrow{ev} A$$

where the first arrow sends $b' \in B'$ to the A-linear operator given by multiplication by b' and the last arrow is the evaluation map. The problem is that the middle arrow, which sends $\lambda' \otimes b'$ to the map $b'' \mapsto \lambda'(b'')b'$, is not an isomorphism. If B' is flat over A, the exact sequences above show that it is an isomorphism and the composition from left to right is the usual trace $\mathrm{Trace}_{B'/A}$. In the general case, we consider the diagram

$$\begin{array}{ccccc} & & \mathrm{Hom}_A(B', A) \otimes_A B' & \longrightarrow & \mathrm{Hom}_A(B', A) \otimes_A B'_g \\ & \overset{\psi}{\nearrow} & \downarrow & \nearrow & \downarrow \\ B' & \xrightarrow[\mu]{} & \mathrm{Hom}_A(B', B') & \longrightarrow & \mathrm{Ker}(F_0^\vee \otimes_A B'_g \to F_1^\vee \otimes_A B'_g) \end{array}$$

By flatness of $A \to B'_g$ we see that the right vertical arrow is an isomorphism. Hence we obtain the unadorned dotted arrow. Since $B'_g = \mathrm{colim} \frac{1}{g^n} B'$, since colimits commute with tensor products, and since B' is a finitely presented A-module we can find an $n \geq 0$ and a B'-linear (for right B'-module structure) map $\psi : B' \to \mathrm{Hom}_A(B', A) \otimes_A B'$ whose composition with the left vertical arrow is $g^n\mu$. Composing with ev we obtain an element $ev \circ \psi \in \mathrm{Hom}_A(B', A)$. Then we set

$$\tau = (ev \circ \psi) \otimes g^{-n} \in \mathrm{Hom}_A(B', A) \otimes_{B'} B'_g = \omega_{B'_g/A} = \omega_{B/A}$$

We omit the easy verification that this element does not depend on the choice of n and ψ above.

Let us prove that τ as constructed in the previous paragraph has the desired property in a special case. Namely, say $B' = C' \times D'$ and $g = (f, h)$ where $A \to C'$ flat, D'_h is flat, and f is a unit in C'. To show: τ maps to $\mathrm{Trace}_{C'/A}$ in $\omega_{C'/A}$. In this case we first choose n_D and $\psi_D : D' \to \mathrm{Hom}_A(D', A) \otimes_A D'$ as above for the pair (D', h) and we can let $\psi_C : C' \to \mathrm{Hom}_A(C', A) \otimes_A C' = \mathrm{Hom}_A(C', C')$ be the map seconding $c' \in C'$ to multiplication by c'. Then we take $n = n_D$ and $\psi = (f^{n_D}\psi_C, \psi_D)$ and the desired compatibility is clear because $\mathrm{Trace}_{C'/A} = ev \circ \psi_C$ as remarked above.

To prove the desired property in general, suppose given $A \to A_1$ with A_1 Noetherian and a product decomposition $B'_g \otimes_A A_1 = C \times D$ with $A_1 \to C$ finite. Set $B'_1 = B' \otimes_A A_1$. Then $\mathrm{Spec}(C) \to \mathrm{Spec}(B'_1)$ is an open immersion as $B'_g \otimes_A A_1 = (B'_1)_g$ and the image is closed as $B'_1 \to C$ is finite (since $A_1 \to C$ is finite). Thus $B'_1 = C \times D'$ and $D'_g = D$. We conclude that $B'_1 = C \times D'$ and g over A_1 are as in the previous paragraph. Since formation of the displayed diagram above

commutes with base change, the formation of τ commutes with the base change $A \to A_1$ (details omitted; use the resolution $F_1 \otimes_A A_1 \to F_0 \otimes_A A_1 \to B_1' \to 0$ to see this). Thus the desired compatibility follows from the result of the previous paragraph. $\square$